FACILITY ONBOARDING TEMPLATE

Compound Risk Detection Engine — Add Facility Document

Facility Name: Visakhapatnam Steel Plant (RINL)

Prepared By: K. Suryanarayana

Designation: Deputy General Manager (Safety & Environment)

Date Prepared: 12-Jul-2026

Instructions: Complete every section below as thoroughly as possible. Add extra rows to any table if you have more entries than space provided. Sections left blank will be flagged during AI validation and you will be asked follow-up questions to complete them. Once complete, convert this document to PDF and upload it in the Add Facility screen.

1. Facility Overview

Facility / Plant Name: Visakhapatnam Steel Plant (VSP)

Physical Address: Ukkunagaram, Gajuwaka, Visakhapatnam, Andhra Pradesh - 530032, India

Industry Sector (e.g. Steel, Thermal Power, Chemical/Pharma, Refinery): Integrated Iron & Steel Manufacturing

Operating Company: Rashtriya Ispat Nigam Limited (RINL) -- Navratna CPSE, Ministry of Steel, Government of India

Commissioning Date: August 1992 (first Blast Furnace blown-in: 1990)

Installed Capacity (e.g. 600 MW, 3 MTPA): 7.3 MTPA liquid steel (post-expansion)

List every major department/unit in the facility (add rows as needed):

Department / Unit Name	Primary Function	Approx. Headcount
Coke Ovens & By-Product Plant	Coking coal carbonization, coke & by-product gas production	850
Sinter Plant	Iron ore agglomeration (sintering) for blast furnace feed	620
Blast Furnace Department (BF-1, BF-2)	Hot metal (pig iron) production	780
Steel Melt Shop-1 & 2 (SMS)	LD converter steelmaking, ladle refining, continuous casting	1450
Light & Medium Merchant Mill / Wire Rod Mill	Hot rolling of TMT bars, structurals, wire rod	1100
Raw Material Handling Plant	Ore/coal yard operations, stacking-reclaiming, logistics	540

Provide a brief narrative overview of what this facility does, end to end:

: Visakhapatnam Steel Plant is India's first major coastal-location integrated steel plant, built to receive

: imported coking coal and export finished steel via the adjacent Visakhapatnam Port. The plant runs a full

: integrated route: coke ovens and a sinter plant feed the blast furnaces, whose hot metal is converted to

: liquid steel in the Steel Melt Shops (SMS-1 and SMS-2) via LD converters, secondary ladle refining, and

: continuous casting. Cast blooms are rolled into TMT rebar, wire rods, and structural sections at the

: Light & Medium Merchant Mill and Wire Rod Mill for domestic construction and infrastructure markets.

2. Process Flow Description

Describe the process step by step, from raw material intake to final product/output. Reference major equipment by name. If a Process Flow Diagram (PFD) or P&ID exists, attach it as an appendix and reference the relevant drawing numbers below.

: 1. Raw material intake: iron ore (Bailadila mines) and imported coking coal received via rail and the

: captive Visakhapatnam Port berth, stacked/reclaimed at the Raw Material Handling Plant.

: 2. Coke Ovens & By-Product Plant: coking coal carbonized into metallurgical coke; by-product gases

: (coke oven gas) recovered and reused as plant fuel.

: 3. Sinter Plant: fine iron ore agglomerated into sinter for improved blast furnace burden permeability.

: 4. Blast Furnace-1 / Blast Furnace-2: sinter, coke, and flux charged from the top; hot blast air injected;

: molten hot metal (pig iron) tapped and transported by torpedo ladle car to the Steel Melt Shop.

: 5. Steel Melt Shop-1 & 2: hot metal charged into LD (oxygen) converters, refined, tapped into ladles,

: secondary-refined, then continuously cast at Casters 1-6 into blooms (Caster-2 handles a portion of

: this casting load).

6. Rolling: cast blooms reheated and hot-rolled at the LMMM and Wire Rod Mill into finished TMT bars,

Drawing Number(s): PFD-VSP-SMS1-014 Rev C; PID-VSP-CASTER2-007 Rev B

3. SCADA / DCS Systems

List every SCADA (Supervisory Control and Data Acquisition) or DCS (Distributed Control System) installed at this facility.

System Name	Vendor	Version	Primary Function	Redundant? (Y/N)
SMS Level-2 Process Automation	ABB Ability	v8.2	Converter blow-end prediction, ladle/caster sequencing	Y
Blast Furnace Control System (BFCS)	Siemens	PCS7 v9.0	Burden charging, hot blast, stove control	Y
Rolling Mills SCADA/HMI	Honeywell Experion PKS	R510	Drive control, pass scheduling, mill sequencing	N

Historian System (if any): PI Historian (OSIsoft/AVEVA), plant-wide, 5-year retention

Network Architecture Notes (e.g. air-gapped, IT/OT segregation): IT/OT segregated via firewalled DMZ; Level-2 automation air-gapped from corporate network, one-way historian data diode to PI

4. Sensor & Instrumentation Details

List every sensor/instrument that should be monitored by the risk detection platform. Add as many rows as needed — this is the single most important table in this document.

Tag ID	Instrument Name	Location	Type	Unit	Normal Range	Alarm Threshold
TE-SMS1-019	Ladle Refractory Temperature	SMS-1, Ladle Bay (Ladle-19)	Thermocouple (Type S)	degC	1580-1650	>1680
PT-BF1-011	Blast Furnace Hot Blast Pressure	Blast Furnace-1, Bustle Main	Pressure Transmitter	bar	3.2-4.0	>4.3
FT-OXY-004	Oxygen Flow to Converter	LD Converter Shop, O2 Lance	Flow Transmitter	Nm3/hr	12000-16000	>17000
LT-CR2-002	Overhead Crane-2 Load Cell	SMS-1, Caster-2 Crane Bay	Load Cell	tonnes	0-180	>185
GT-COB-007	Coke Oven Gas (CO) Detector	Coke Oven Battery-2, Ascension Pipe	Gas Detector (Electrochemical)	ppm	0-30	>50
TE-CAST2-003	Tundish Temperature, Caster-2	SMS-1, Caster-2 Tundish	Thermocouple (Type B)	degC	1520-1560	>1580
VT-CR2-008	Overhead Crane-2 Hoist Vibration	SMS-1, Caster-2 Crane Bay	Vibration Sensor	mm/s	0-4.5	>7.1
PT-COMP-012	Compressed Air Header Pressure	Central Compressor House	Pressure Transmitter	bar	6.0-7.5	<5.0

4.1 Real-Time Data Integration (API Access)

If sensor readings are available via a JSON API (recommended), provide connection details below for each system. If not yet available, leave blank — you can configure this later in the portal.

API Base URL: https://scada-gateway.vsp-internal.example/api/v1/tags

Authentication Method (e.g. API key, Bearer token, none): API key (rotated quarterly)

Polling Frequency (e.g. every 30 seconds): Every 15 seconds (critical tags); every 60 seconds (utility tags)

5. Key Personnel

5.1 Managerial Staff

Name	Designation	Department	Email	Phone
K. Suryanarayana	Deputy General Manager, Safety & Environment	Safety & Environment	k.suryanarayana@vsp-rinl.example	+91-891-278-1021
P. Ramesh Babu	General Manager, Works	Works (Overall Operations)	p.rameshbabu@vsp-rinl.example	+91-891-278-1002
T. Venkata Rao	Deputy General Manager, SMS	Steel Melt Shop-1 & 2	t.venkatarao@vsp-rinl.example	+91-891-278-1108
N. Satish Kumar	General Manager, Blast Furnace	Blast Furnace Department	n.satishkumar@vsp-rinl.example	+91-891-278-1155
A. Lakshmi Prasanna	Deputy General Manager, HR & Contract Labour	Human Resources	a.lakshmiprasanna@vsp-rinl.example	+91-891-278-1210
S. Chandrasekhar	Manager, Maintenance Planning	Central Maintenance	s.chandrasekhar@vsp-rinl.example	+91-891-278-1244

5.2 Maintenance Staff

Name	Role	Shift	Email	Phone	Certifications
B. Nageswara Rao	Senior Foreman, Mechanical Maintenance	A	b.nageswararao@vsp-rinl.example	+91-98480-11234	NEBOSH IGC, Rigging & Lifting Cert.
Y. Srinivas	Electrical Maintenance Engineer	B	y.srinivas@vsp-rinl.example	+91-98480-11897	Electrical Safety (HV), CEA Cert.
D. Krishna Murthy	Crane & Hoist Maintenance Supervisor	A	d.krishnamurthy@vsp-rinl.example	+91-98480-12045	Crane Inspector Certification
M. Appalanaidu	Refractory Maintenance Technician	C	m.appalanaidu@vsp-rinl.example	+91-98480-12310	Refractory Installation & Inspection Cert.
R. Ganesh	Instrumentation Maintenance Engineer	B	r.ganesh@vsp-rinl.example	+91-98480-12588	Instrumentation & Control Systems Cert.
V. Bhaskara Rao	Hydraulics & Pneumatics Technician	A	v.bhaskararao@vsp-rinl.example	+91-98480-12799	Hydraulic Systems Maintenance Cert.

5.3 Safety Officers

Name	Role	Email	Phone
K. Suryanarayana	Chief Safety Officer	k.suryanarayana@vsp-rinl.example	+91-891-278-1021
J. Padmavathi	Safety Officer, SMS-1 & 2	j.padmavathi@vsp-rinl.example	+91-98480-13120
C. Ravindra Babu	Safety Officer, Blast Furnace & Coke Ovens	c.ravindrababu@vsp-rinl.example	+91-98480-13355
S. Vijaya Lakshmi	Fire & Emergency Response Officer	s.vijayalakshmi@vsp-rinl.example	+91-98480-13590

5.4 Escalation Logic

Describe who is contacted, in what order, for each severity of incident. Add rows for as many escalation levels as your facility uses.

Level	Any abnormal reading, near-miss, or minor incident on shift	Shift Engineer / Area Foreman	Immediate (on detection)
1 — First Responder	Injury, equipment damage, or confirmed alarm breach	Deputy GM (Area) + Safety Officer	Within 15 minutes
2 — Shift Engineer	Fatality risk, fire, explosion, or multi-area impact	GM (Works) + Chief Safety Officer	Within 5 minutes
3 — Plant Head	Fatality, major fire, or regulatory-reportable event	CMD / Corporate Office + Steel Ministry liaison + APPCB/DGFASLI	Within 30 minutes
4 — Corporate / Regulatory			

6. Shift & Workforce Patterns

List all shift timings observed at this facility:

Shift Name	06:00	14:00	~1450 (across SMS/BF/Rolling)
Shift A	14:00	22:00	~1450 (across SMS/BF/Rolling)
Shift B	22:00	06:00	~1050 (reduced night complement)
Shift C			

Describe the shift changeover / handover protocol — what is checked, communicated, and signed off between outgoing and incoming shifts:

: Outgoing shift-in-charge briefs incoming counterpart on: equipment status/any open work permits,

: abnormal readings or alarms in the last 8 hours, deferred maintenance items still pending closure,

: and any near-miss or safety observation raised during the shift. Both sign the shift log register

: and the crane/ladle pre-use checklist before handover is considered complete. SMS-1 additionally

: requires a joint walk-down of the casting bay (Caster-1 through Caster-6) before sign-off.

: ____

7. Maintenance Records & Timelines

List major equipment and their maintenance history. Explicitly note any deferred or overdue maintenance items — this is critical information for risk detection.

Equipment	Last Maintenance Date	Type	Next Scheduled	Performed By	Deferred? Notes
Overhead Crane-2 Hoist Brake System (SMS-1)	18-Feb-2026	Preventive	18-Aug-2026	Central Maintenance (Mechanical)	DEFERRED -- full brake overhaul postponed from Mar-2026 due to production schedule; interim inspection only completed
Ladle Refractory Lining, Ladle-19	02-May-2026	Relining	02-Sep-2026	Refractory Maintenance Team	On schedule
Blast Furnace-1 Hot Blast Stove Refractory	10-Jan-2026	Preventive	10-Jul-2026	Central Maintenance (Mechanical)	On schedule
LD Converter No.2 Vessel Trunnion	22-Nov-2025	Preventive	22-May-2026	Central Maintenance (Mechanical)	On schedule
Coke Oven Battery-2 Gas Collection Main	05-Apr-2026	Corrective (gasket replacement)	05-Oct-2026	Coke Ovens Maintenance	On schedule
Compressed Air Header, Central Compressor House	14-Mar-2026	Preventive	14-Sep-2026	Utilities Maintenance	On schedule

8. Incident & Negligence History

List any prior incidents, near-misses, safety violations, or regulatory citations at this facility, even if minor. Patterns of recurrence are especially important for risk modeling.

Date	Incident Description	Root Cause (if known)	Corrective Action Taken	Recurred? (Y/N)
08-Jun-2026	Sudden explosion at Caster-2, Steel Melt Shop-1 during casting (~16:15 hrs), before slide-gate opening to pour molten steel from ladle to tundish. Fireball rose to ceiling, molten steel spilled, Overhead Crane-2 caught fire. 8 workers killed, 6 injured.	Under investigation -- 3-member external inquiry (Steel Ministry, headed by Director-in-Charge, Bokaro Steel Plant) assessing ladle/slide-gate integrity and possible entrapped-gas ignition; deferred crane brake overhaul (see Maintenance table) also under review as a contributing factor	Plant-wide ladle & crane inspection ordered; casting operations under enhanced review across Casters 1-6; revised exclusion-zone protocol during pouring	Y
08-Jun-2012	Explosion during commissioning of Oxygen (LD) Converter No.1; 19 workers killed	Entrapped gas ignition during commissioning trials, per High Level Committee inquiry	Revised commissioning specification for oxygen-enriched systems; fire-prevention training conducted with IIT Kharagpur	Y
19-Sep-2025	Minor coke oven gas (CO) leak near ascension pipe, Coke Oven Battery-2 -- no injuries, area evacuated as precaution	Corroded gasket at flange joint	Gasket replaced; inspection frequency increased to monthly for that joint class	N
03-Jan-2026	Wire rope near-miss (partial strand failure detected) on Raw Material Handling Plant stacker-reclaimer crane	Wear beyond service limit, inspection interval too long	Wire rope replaced; NDT inspection interval shortened from 6 months to 3 months plant-wide for yard cranes	N

Is there any broader history of negligence (e.g. deferred safety audits, understaffing, ignored worker complaints) that may make this facility more susceptible to future incidents? Describe below:

: Worker representatives and trade unions (CITU) have on record alleged long-standing ageing-equipment

: and manpower-shortage issues across SMS-1, and have called for a comprehensive, plant-wide safety audit

: following the 8-Jun-2026 Caster-2 ladle explosion. This is the second fatal casting/converter-area

: incident in the plant's history (see 2012 entry below), which external commentators have flagged as

: evidence of a recurring pattern rather than an isolated failure. A judicial inquiry and a separate

: 3-member external technical committee (headed by the Director-in-Charge, Bokaro Steel Plant) are both underway.

9. Employee Directory

Provide full details for all on-site employees relevant to safety operations. This information is used for shift-correlation analysis and emergency response — please complete every column.

Name	Role / Dept.	Email	Phone	Blood Group	Working Hours	Working Days	Emergency Contact (Name / Phone / Relation)
G. Ravi Teja	Senior Operator, SMS-1 Caster Bay	g.raviteja@vsp-rinl.example	+91-98480-21001	O+	06:00-14:00	Mon-Sat	G. Lakshmi (Spouse) / +91-98480-91001
M. Suresh	Crane Operator, SMS-1	m.suresh@vsp-rinl.example	+91-98480-21045	B+	14:00-22:00	Mon-Sat	M. Padma (Spouse) / +91-98480-91045
K. Anitha	Shift Engineer, Blast Furnace-1	k.anitha@vsp-rinl.example	+91-98480-21089	A+	22:00-06:00	Mon-Sat	K. Rao (Father) / +91-98480-91089
P. Durga Rao	Ladle Bay Technician, SMS-1	p.durgarao@vsp-rinl.example	+91-98480-21133	AB+	06:00-14:00	Mon-Sat	P. Sita (Spouse) / +91-98480-91133
V. Naga Malleswari	Control Room Operator, LMMM	v.nagamalleswari@vsp-rinl.example	+91-98480-21177	O-	14:00-22:00	Mon-Sat	V. Rajesh (Spouse) / +91-98480-91177
S. Praveen Kumar	Fire & Safety Technician	s.praveenkumar@vsp-rinl.example	+91-98480-21221	B-	Rotating (24x7 cover)	6-day rotating roster	S. Vani (Spouse) / +91-98480-91221
N. Vara Lakshmi	Instrumentation Technician, SMS-2	n.varalakshmi@vsp-rinl.example	+91-98480-21265	A-	06:00-14:00	Mon-Sat	N. Ramana (Father) / +91-98480-91265
T. Bhargav	Coke Oven Battery Operator	t.bhargav@vsp-rinl.example	+91-98480-21309	O+	22:00-06:00	Mon-Sat	T. Kumari (Mother) / +91-98480-91309
R. Sai Kiran	Raw Material Handling Plant Operator	r.saikiran@vsp-rinl.example	+91-98480-21353	AB-	14:00-22:00	Mon-Sat	R. Devi (Spouse) / +91-98480-91353
L. Manikyala Rao	Maintenance Fitter, Central Maintenance	l.manikyalarao@vsp-rinl.example	+91-98480-21397	B+	06:00-14:00	Mon-Sat	L. Kanaka Durga (Spouse) / +91-98480-91397

10. Attendance & Access Control Systems

System Name / Vendor: Matrix COSEC (biometric + RFID card)

API Base URL (if available): https://attendance.vsp-internal.example/api/v2/logs

Authentication Method: Bearer token

Data Format (e.g. JSON, CSV export): JSON

11. Utility & Support Systems

List critical utility systems that support plant operations. Failures in these systems often compound with primary process risks (e.g. a power backup failure during a sensor fault).

Diesel Generator (4x2.5 MVA) + Central UPS	**N+1 redundant DG bank, 30-min UPS ride-through**	**05-Jun-2026**	**Last Tested Date**
Central Compressor House (screw compressors)	N+1 redundant compressors	14-Mar-2026	
Cooling Water Recirculation System	Dual pump sets, standby header	20-Apr-2026	
Plant-wide Hydrant + Foam/CO2 (oil cellars)	Dual fire water pumps (electric + diesel-driven jockey)	22-May-2026	
Central Control Room HVAC	N+1 redundant AHUs	01-Jun-2026	
HVAC (Control Room / Critical Areas)			

12. Training & Certification Records

Summarize the safety training program in place. Individual employee certifications should already be captured in Section 9; this section covers program-level information.

Safety Induction Program (frequency, duration): 3-day induction for all new joinees + contract workers; refresher every 12 months

Mock Drill Frequency (fire, evacuation, gas leak, etc.): Quarterly (fire); half-yearly (evacuation); half-yearly (gas leak / H2S & CO)

Date of Last Mock Drill: 22-May-2026 (fire mock drill, SMS-1)

Contractor / Third-Party Worker Safety Training Process: Mandatory 1-day safety orientation + PPE certification before contractor gate-pass issuance; renewed annually

13. Environmental & Quality Compliance

Environmental Clearance / Consent to Operate (reference number): APPCB/CFO/VSP/2024/00417 (Consent to Operate)

Pollution Control Board Registration: Andhra Pradesh Pollution Control Board Reg. No. APPCB-VSKP-RINL-0091

Quality Management System Certification (e.g. ISO 9001, if any): ISO 9001:2015

Environmental Management System Certification (e.g. ISO 14001, if any): ISO 14001:2015

Occupational Health & Safety Certification (e.g. ISO 45001, if any): ISO 45001:2018

14. Emergency Response & Compliance

List applicable regulatory standards this facility operates under:

Regulatory Standards (e.g. Factory Act, OISD, DGMS, CEA guidelines): Factories Act 1948; OISD-STD-192 (fire/process safety); CEA guidelines; APPCB consent conditions

Fire Safety Systems Present (hydrant, sprinkler, detection, etc.): Hydrant network (plant-wide), foam/CO2 systems (oil cellars), automatic sprinklers (cable galleries), fixed gas detection (coke ovens)

Date of Last Safety Audit: 30-Mar-2026 (internal); external Steel Ministry probe ongoing post 8-Jun-2026 incident

Summarize key findings from the last safety audit, if any:

: Last internal audit (Mar-2026) flagged deferred crane brake/hoist inspections in SMS-1 and recommended

: closure within 60 days -- not fully closed at time of the 8-Jun-2026 incident. External Steel Ministry

: inquiry (ongoing) is separately assessing ladle/slide-gate integrity and casting-bay staffing levels.

: ____

Declaration

I confirm that the information provided in this document is accurate and complete to the best of my knowledge at the time of submission.

Prepared By (Name & Signature): K. Suryanarayana

Reviewed By (Name & Signature): P. Ramesh Babu, General Manager (Works)

Date: 12-Jul-2026